Section 2.5

7. a) $f(x)$ is not defined at $x=2$.
   b) $f(x)$ is not continuous at $x=2$

8. $f(x)$ is continuous at $[-1,0)\cup(0,1)\cup(1,2)\cup(2,3)$

9. $f(2)$ should be equal to 0 to let $f(x)$ continuous at $x=2$

10. $f(1)$ should be equal to 2 to remove the discontinuity.

26. Every point in $[\frac{1}{3},\infty)$ make the function $y=f(x)$ continuous.

27. $y=(2x-1)^{1/3}$ is continuous on every real number.

32. Continuous

38. Not continuous

43. $a=\frac{4}{3}$

47. $a=\frac{5}{2}$, $b=\frac{1}{2}$

55. Let $y=f(x)=x^3-15x$ (The solutions of $x^3-15x+1=0$ satisfies $f(x)=-1$)

$f(-4)=(-4)^3-15(-4)=-4$ }
$f(-3)=(-3)^3-15(-3)=18$ } $\rightarrow$ By IVT, there is x_0 between -4 and -3 such that $f(x_0)=-1$

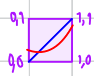
$f(0)=0^3-15(0)=0$ }
$f(1)=1^3-15(1)=-14$ } $\rightarrow$ By IVT, there is x_1 between 0 and 1 such that $f(x_1)=-1$

$f(4)=4^3-15(4)=4$ } $\rightarrow$ By IVT, there is x_2 between 1 and 4 such that $f(x_2)=-1$

Hence, the equation $x^3-15x+1=0$ has three solutions in the interval $[-4,4]$

65. True. By IVT, if the function changes sign on an interval and the function is continuous on that interval, it means that the graph of the function will pass zero at least once on that interval. Hence it contradicts with the fact that the function is never zero on the interval.

67. Let $g(x)=f(x)-x \rightarrow g(x)$ is a continuous function and $-1\le g(x)\le 1$ for every $x\in[0,1]$
Since $g(0)=f(0)-0\ge 0$ and $g(1)=f(1)-1\le 0$.
By IVT, there exist x_0 in $[0,1]$ such that $g(x_0)=0$ or $f(x_0)=x_0$.



70. If $\lim_{h\to 0} f(c+h)=f(c)$, then f is continuous at c.
Let $f(x)=\sin x$. $\lim_{h\to 0} f(c+h)=\lim_{h\to 0}\sin(c+h)=\lim_{h\to 0}\sin c\cos h+\sin h\cos c=\sin c\cdot 1+0\cdot\cos c=\sin c=f(c)$
$\therefore f(x)=\sin x$ is continuous at every point $x=c$.
Let $g(x)=\cos x$, $\lim_{h\to 0} g(c+h)=\lim_{h\to 0}\cos(c+h)=\lim_{h\to 0}\cos c\cos h-\sin c\sin h=\cos c\cdot 1-\sin c\cdot 0=\cos c=g(c)$
$\therefore g(x)=\cos x$ is continuous at every point $x=c$.